Information Technology Institute

ITI

Introduction to Computer Networks & Cyber Security

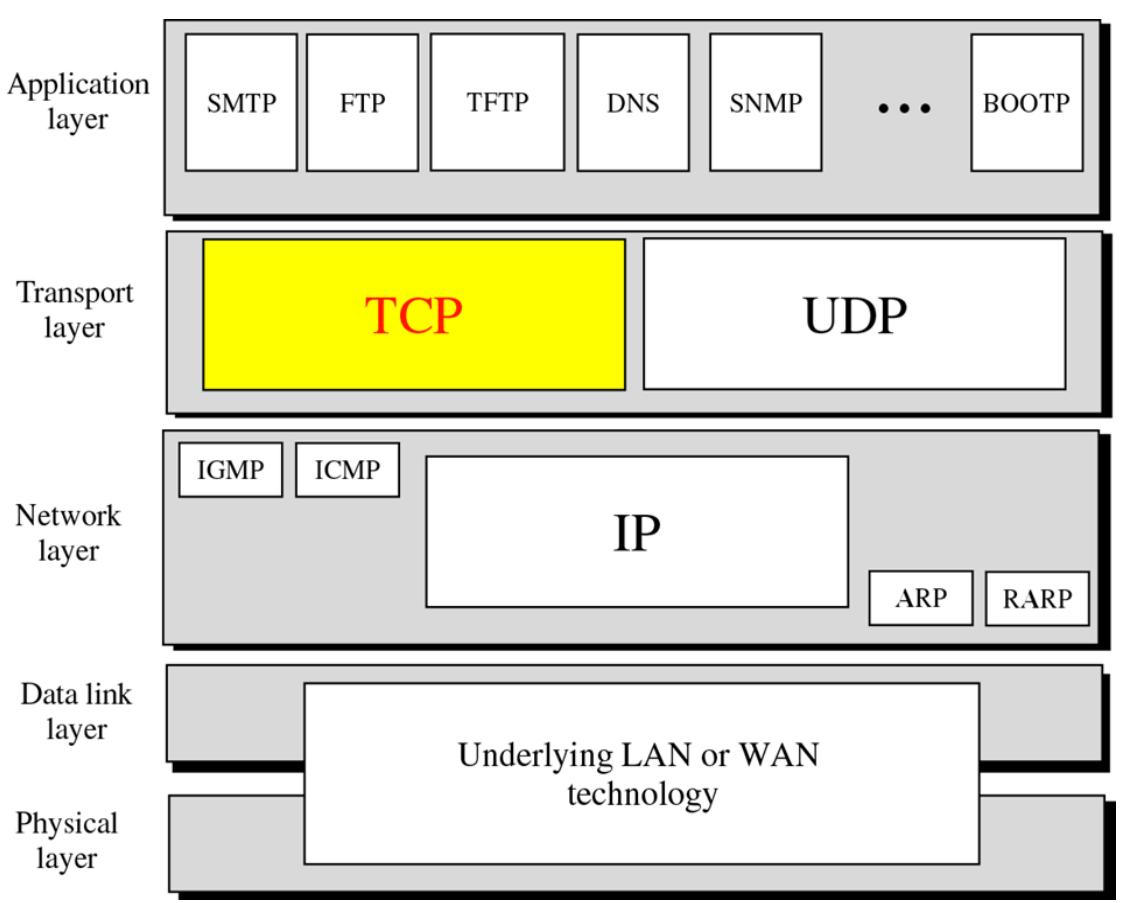
Prepared By : Mohamed AboSehly

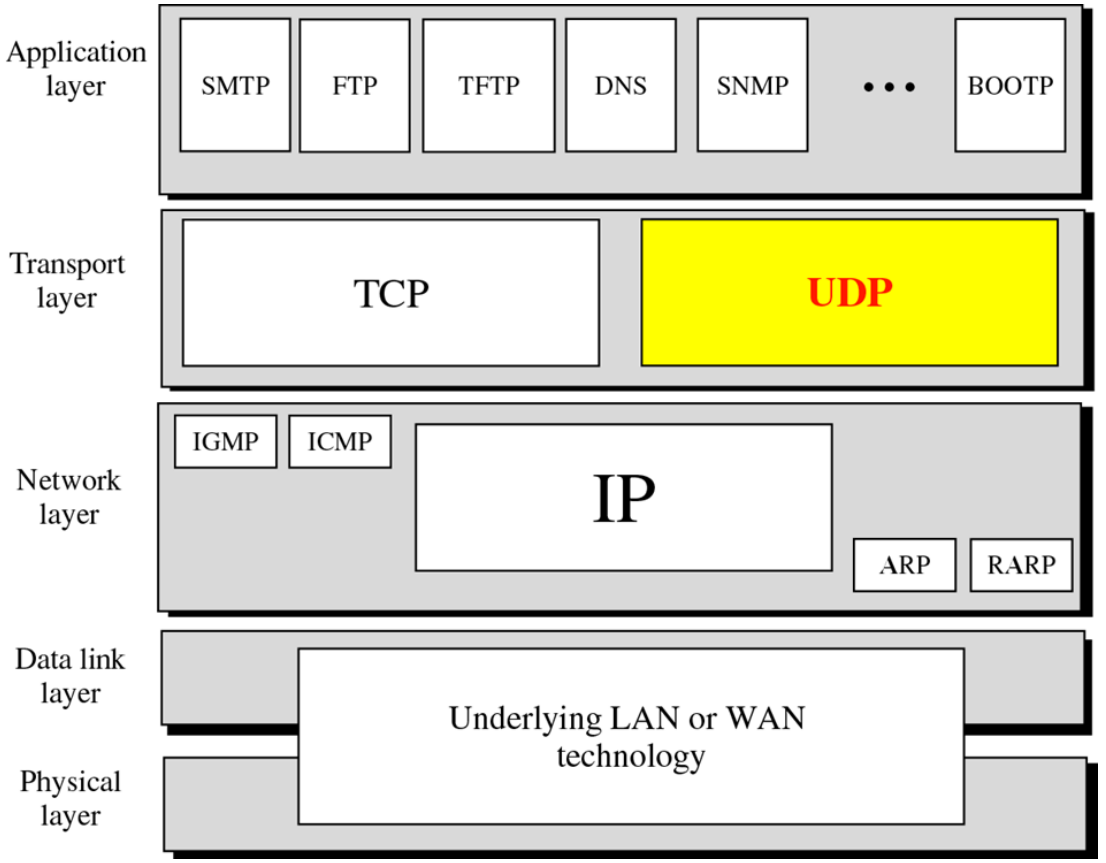
TCP/IP Protocol Architecture

Transport Layer

TCP/IP Protocol Architecture

TCP/IP Protocol Architecture





TCP/IP Protocol Architecture

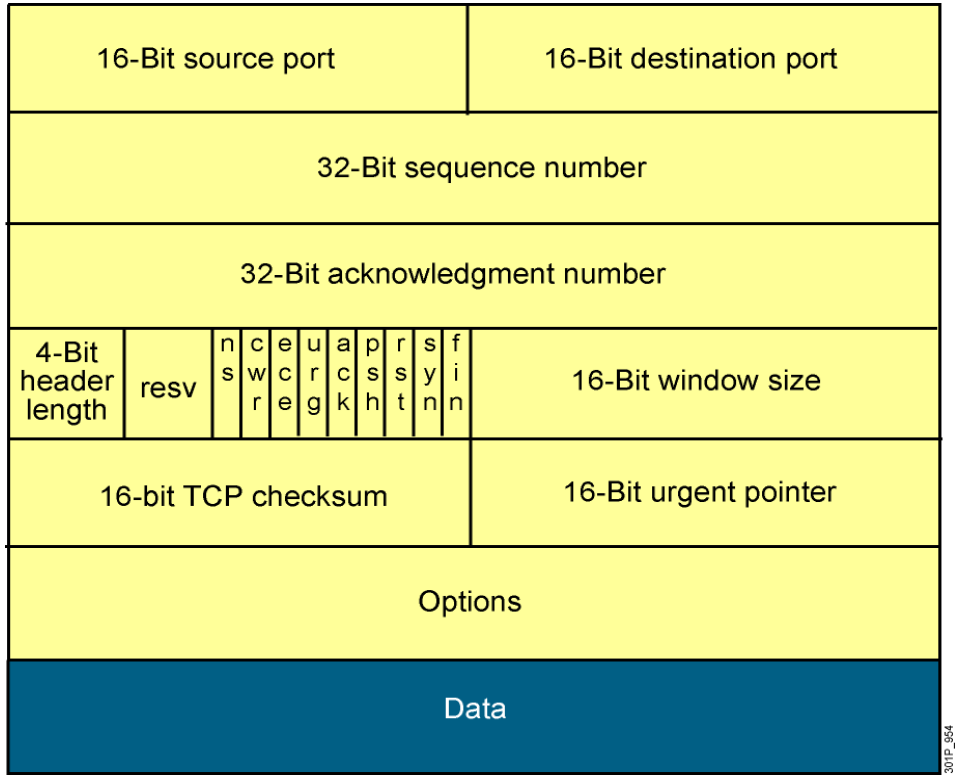
TCP Characteristics

- Transmission Control Protocol
- Transport layer protocol
- Use port numbers
- Reliable (Acknowledgement of receipt)
- **Connection oriented(**synchronization)
- **Full duplex**
- Error control(Error checking(checksum)
- **Flow control**
  - Data-recovery features
  - Sequencing of data packets



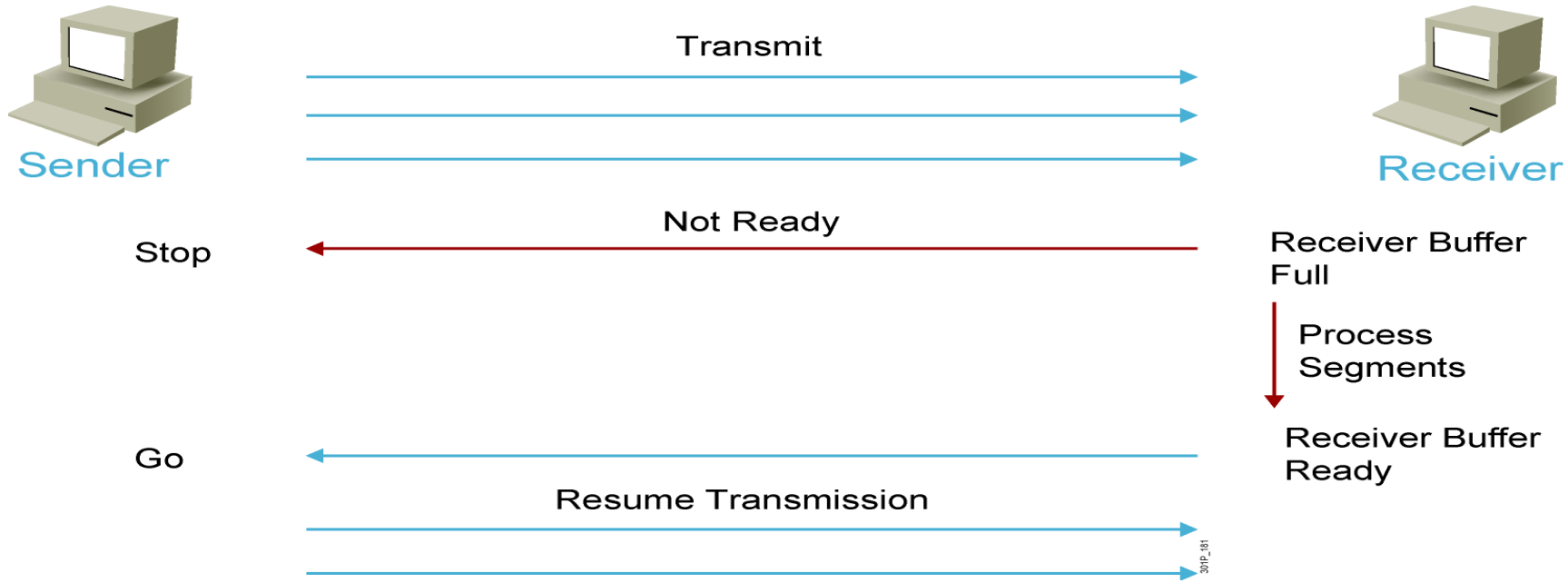
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TCP Header



TCP/IP Protocol Architecture

Flow Control

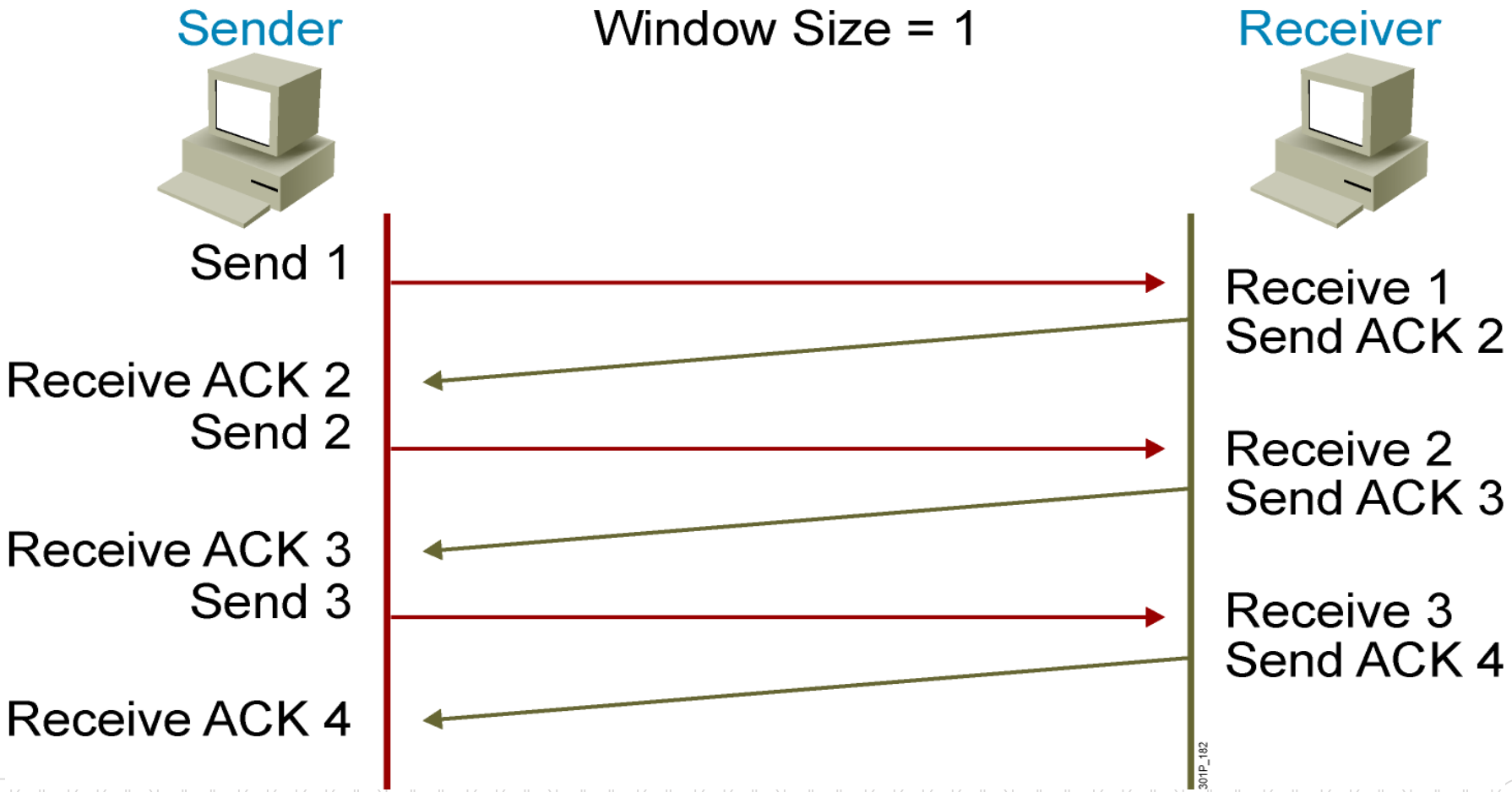


Flow Control:

If the transmitter is sending data faster than the receiver so the receiver will drop the data and the retransmitting will waste time and network resources. The Round trip time will be very slow so we used **TCP windowing**

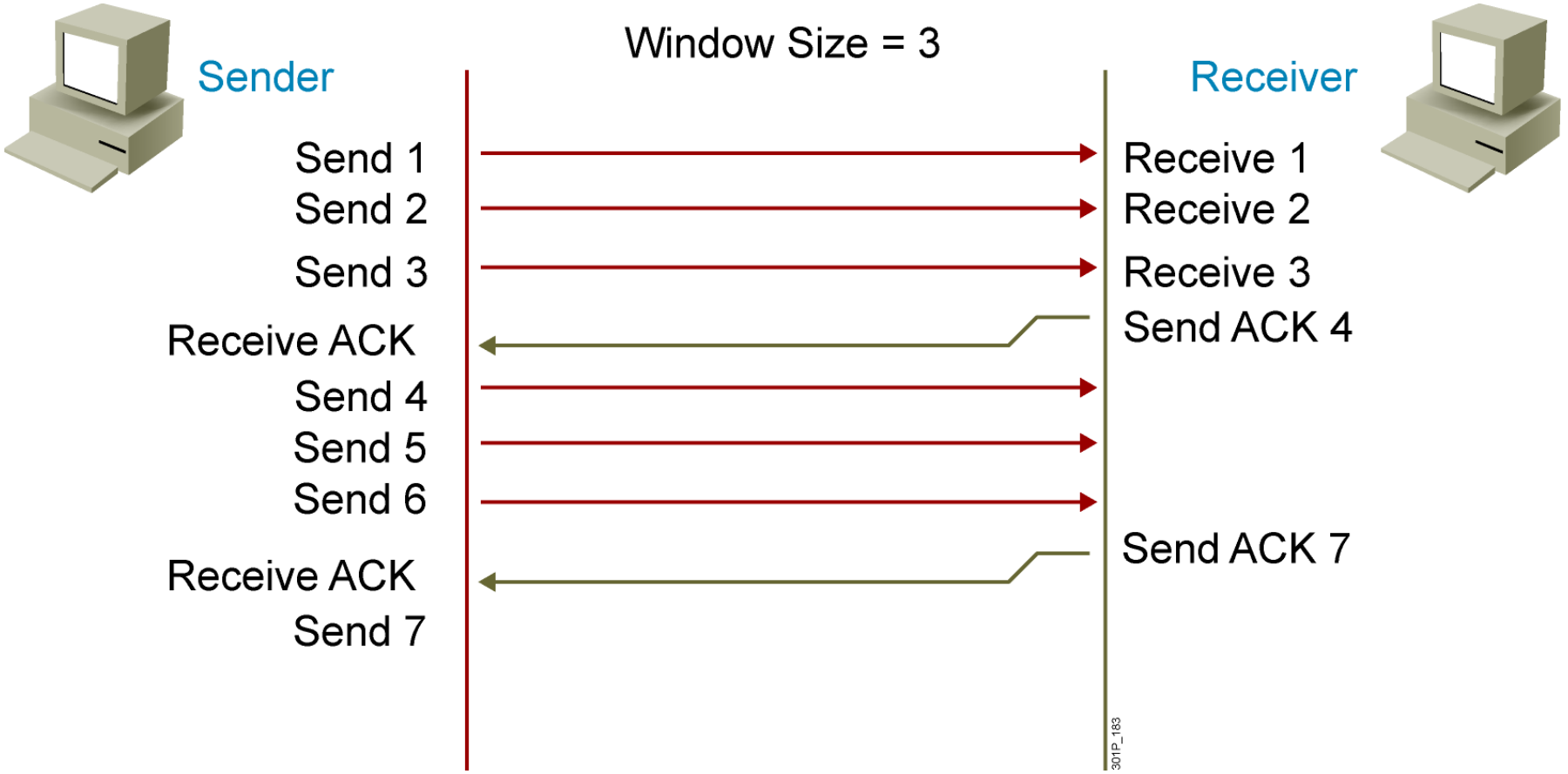
TCP/IP Protocol Architecture

TCP Acknowledgment



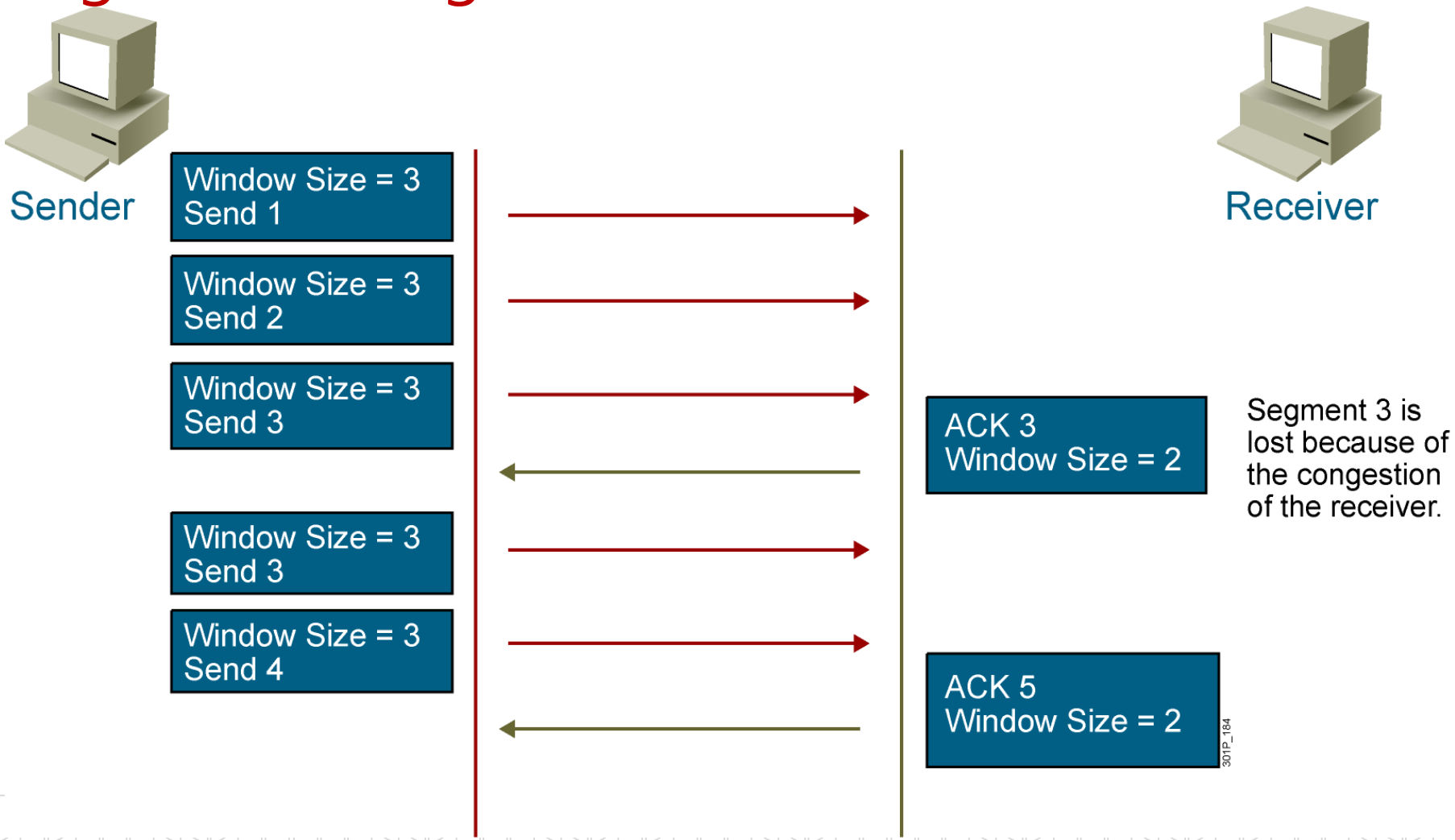
TCP/IP Protocol Architecture

Fixed Windowing



TCP/IP Protocol Architecture

TCP Sliding Windowing

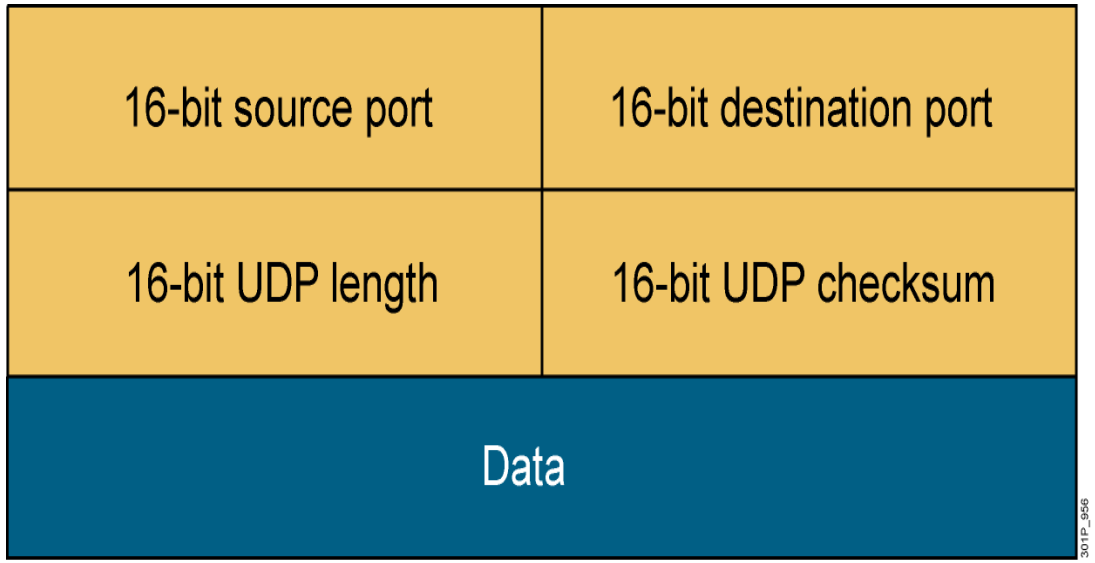


TCP/IP Protocol Architecture

UDP Characteristics

- User Datagram Protocol / Transport layer protocol
- Process to process communication
  - Use port numbers
- **Connectionless** (no notification)
- Unreliable
- **Perform very limited error checking**
- Very simple using a minimum of overhead
- Provides best-effort delivery
    - The data may be dropped due to:
        - Routing Error,
        - Duplicate data due to redundancy
        - Data loss in its way due to TTL.
- **Has no data-recovery features**

UDP Header



TCP/IP Protocol Architecture

TCP (Reliable) vs. UDP (Best-Effort Comparison)

	Reliable	Best-Effort
Connection Type	Connection-oriented	Connectionless
Protocol	TCP	UDP
Sequencing	Yes	No
Uses	▪ E-mail ▪ File sharing ▪ Downloading	▪ Voice streaming ▪ Video streaming

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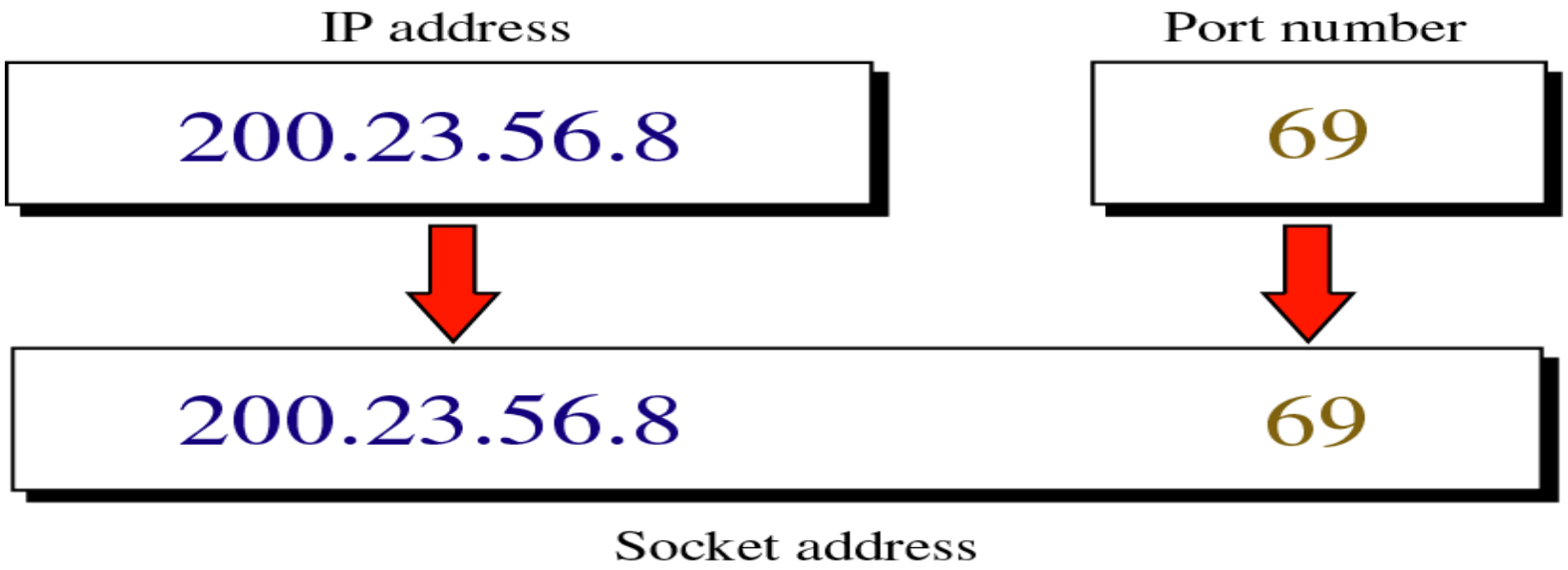
TCP/IP Protocol Architecture

Port Numbers

- **Well-Known Ports (0–1023)_** (Reserved)
  - Reserved for **standard system services** and **widely used** protocols.
  - Examples : **HTTP (port 80), HTTPS (port 443), FTP (port 21), and SSH (port 22).**
- **Registered Ports (1024–49151)_** (Assigned)
  - **Assigned** by ICANN for **specific applications or services**.
  - Used by **software developers** and organizations for applications like **MySQL (port 3306) or SQL Server (port 1433).**
- **Dynamic (49152–65535)** (Private Ports)
  - **Temporarily ports** assigned for **client-side connections**.
  - Commonly used when a device initiates a connection to a server, such as a web browser connecting to a website.
  - **not assigned** and can be **reused for different sessions**
- https://en.wikipedia.org/wiki/List_of_TCP_and_UDP_port_numbers

TCP/IP Protocol Architecture

Socket Address



TCP/IP Protocol Architecture _ Lab

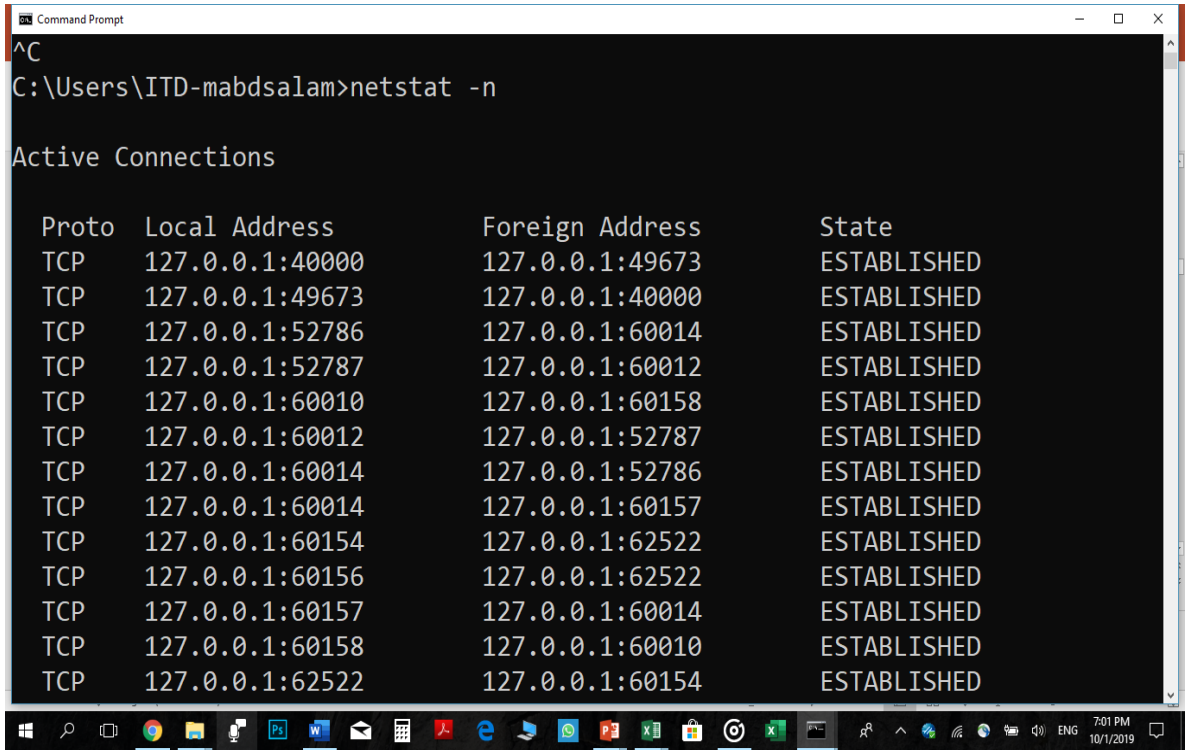
❖ NETSTATE

netstat –n

netstat –a

netstat –o

To know session and ports on your device

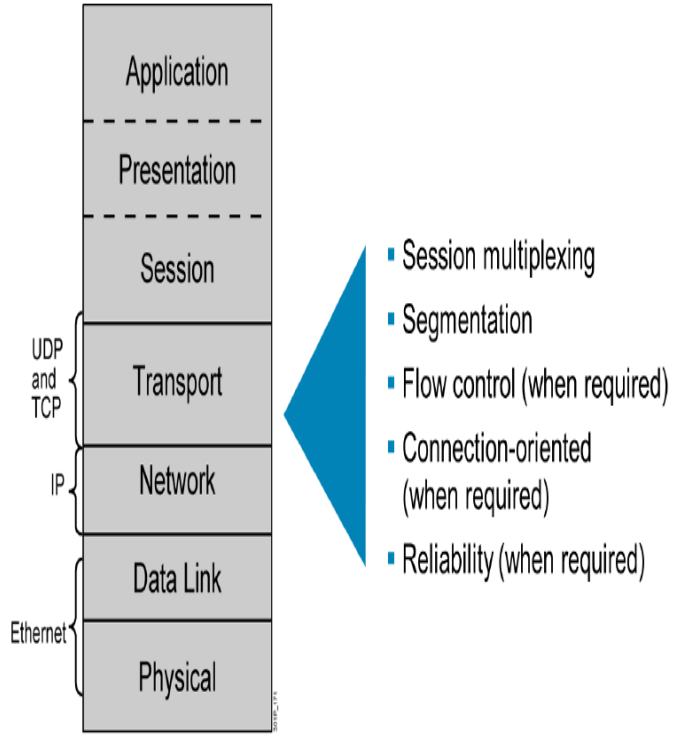


TCP/IP Protocol Architecture

Transport Layer Summery

- **Session multiplexing:**
  - open **multiple sessions** using **UDP and TCP**

  Example : open **cisco.com & open facebook.com** you are the source using **port 49001** and another **port 49002** on the same machine (session multiplexing).
- **Segmentation:**
  - divided the data up to **multiple segments** to be easier in handling (the maximum performance 1518 byte)
- **Connection Oriented:**
  - To maintain the session with **acknowledgements** that the data are sent to the receiver then terminate the session
- **Reliability:**
  - **Data corrections** and **avoid** the **duplicate data** out of order and data arrangement.

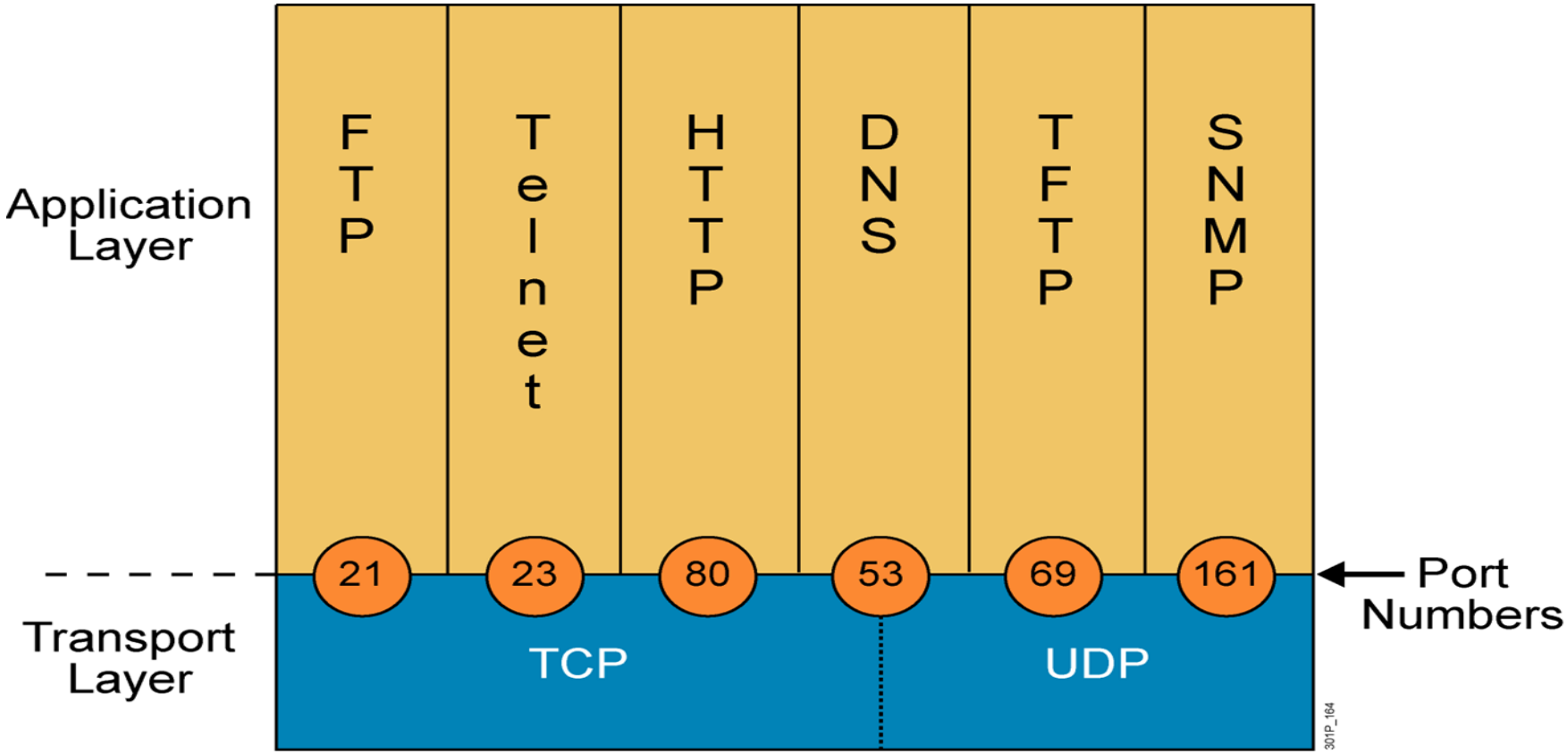


TCP/IP Protocol Architecture

Application Layer

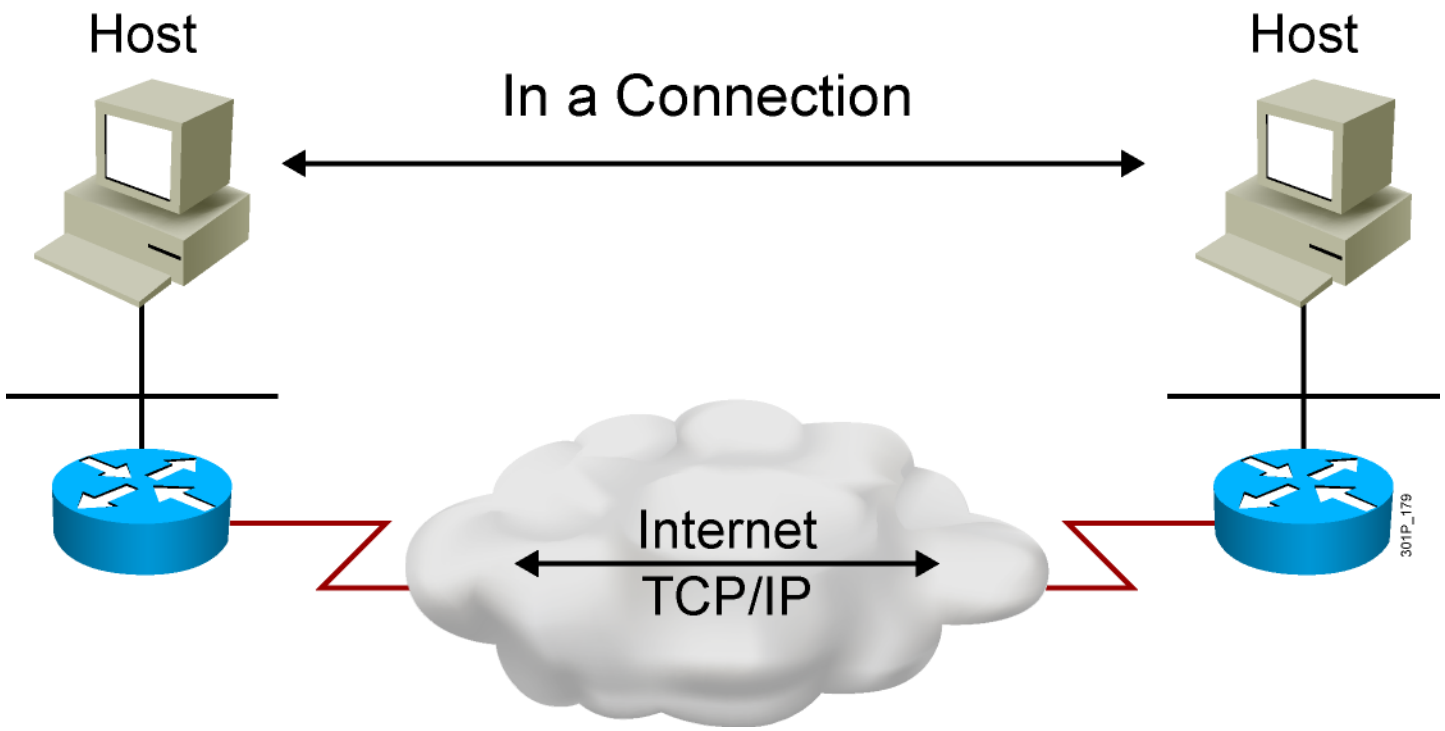
TCP/IP Protocol Architecture

Mapping Layer 4 to Applications



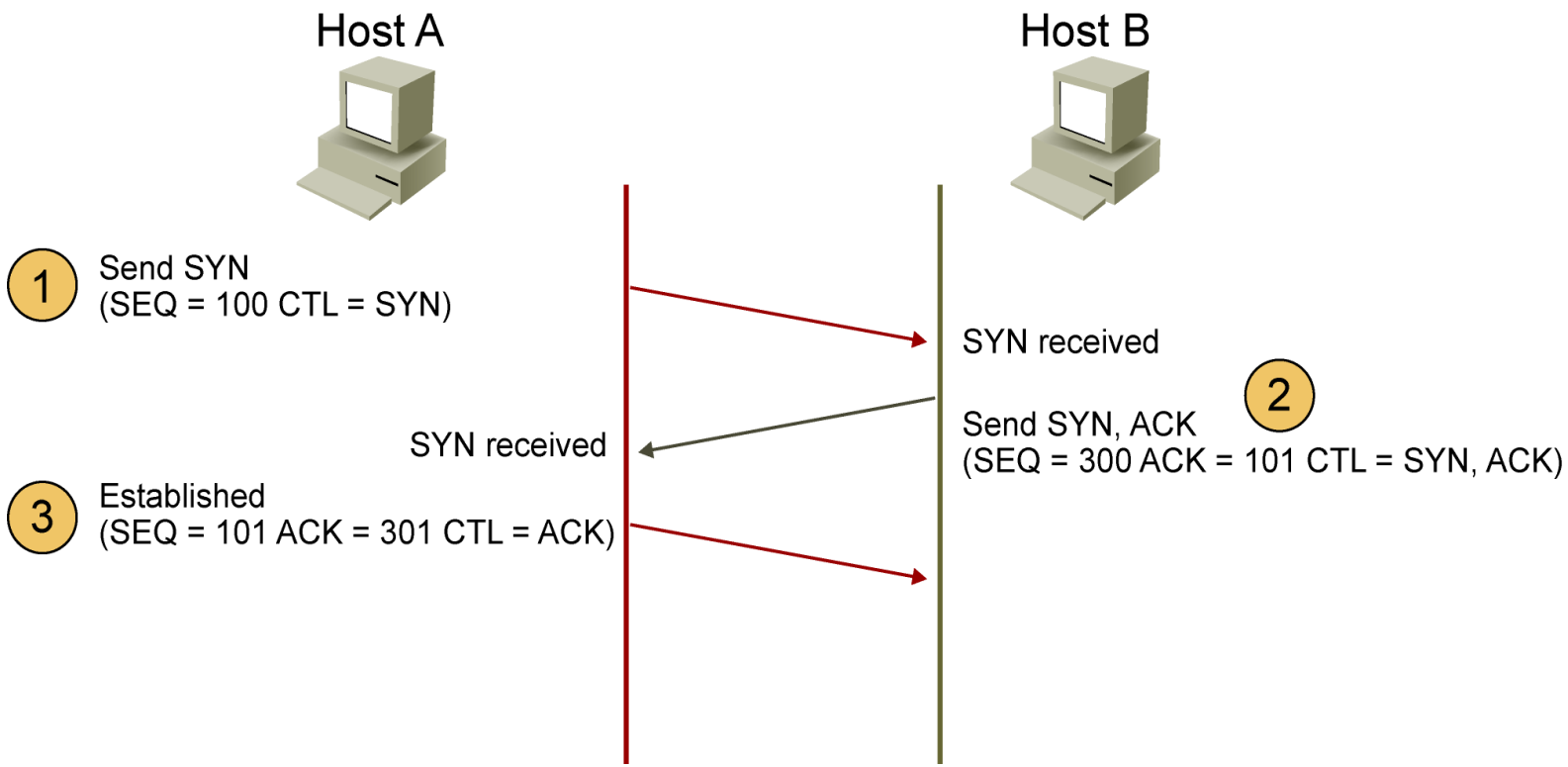
TCP/IP Protocol Architecture

ESTABLISHING A CONNECTION



TCP/IP Protocol Architecture

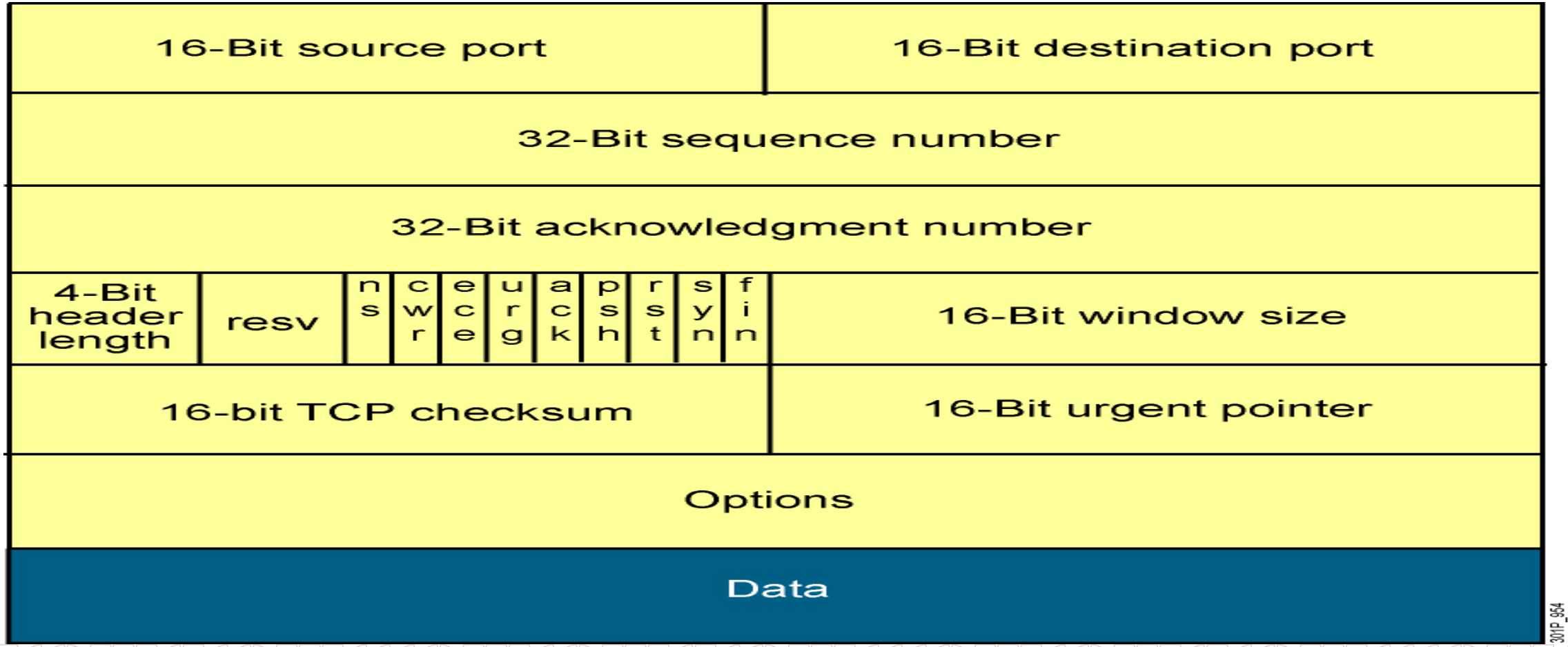
THREE-WAY HANDSHAKE



CTL = Which control bits in the TCP header are set to 1

Part 1 (TCP/IP Protocol Architecture)

TCP



Thank You